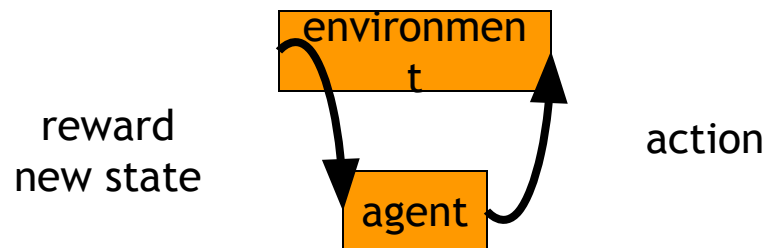


Reinforcement Learning Tutorial

Peter Bodík
RAD Lab, UC Berkeley

Previous Lectures

- Supervised learning
 - classification, regression
- Unsupervised learning
 - clustering
- Reinforcement learning
 - more general than supervised/unsupervised learning
 - learn from interaction w/ environment to achieve a goal



Today

- examples
- defining an RL problem
 - Markov Decision Processes
- solving an RL problem
 - Dynamic Programming
 - Monte Carlo methods
 - Temporal-Difference learning

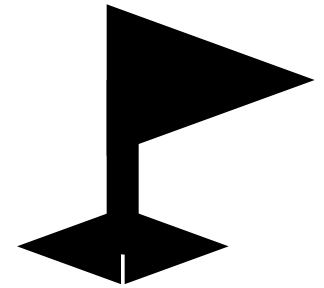
Robot in a room

			+1
			-1
START			

actions: UP, DOWN, LEFT, RIGHT

UP

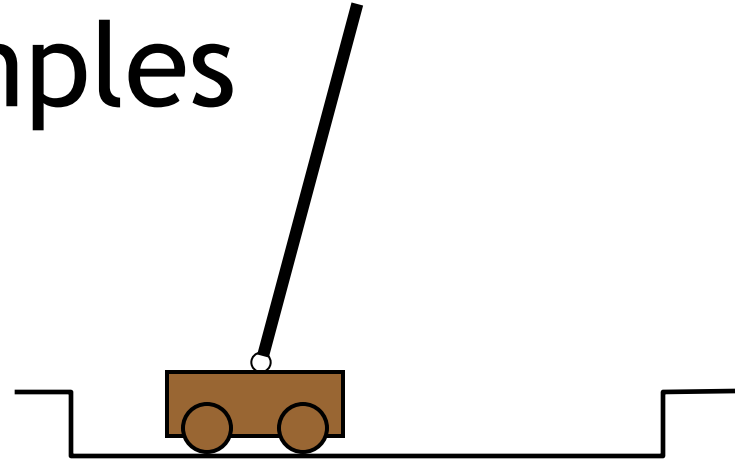
80% move UP
10% move LEFT
10% move RIGHT



- reward +1 at [4,3], -1 at [4,2]
- reward -0.04 for each step
- what's the strategy to achieve max reward?
- what if the actions were deterministic?

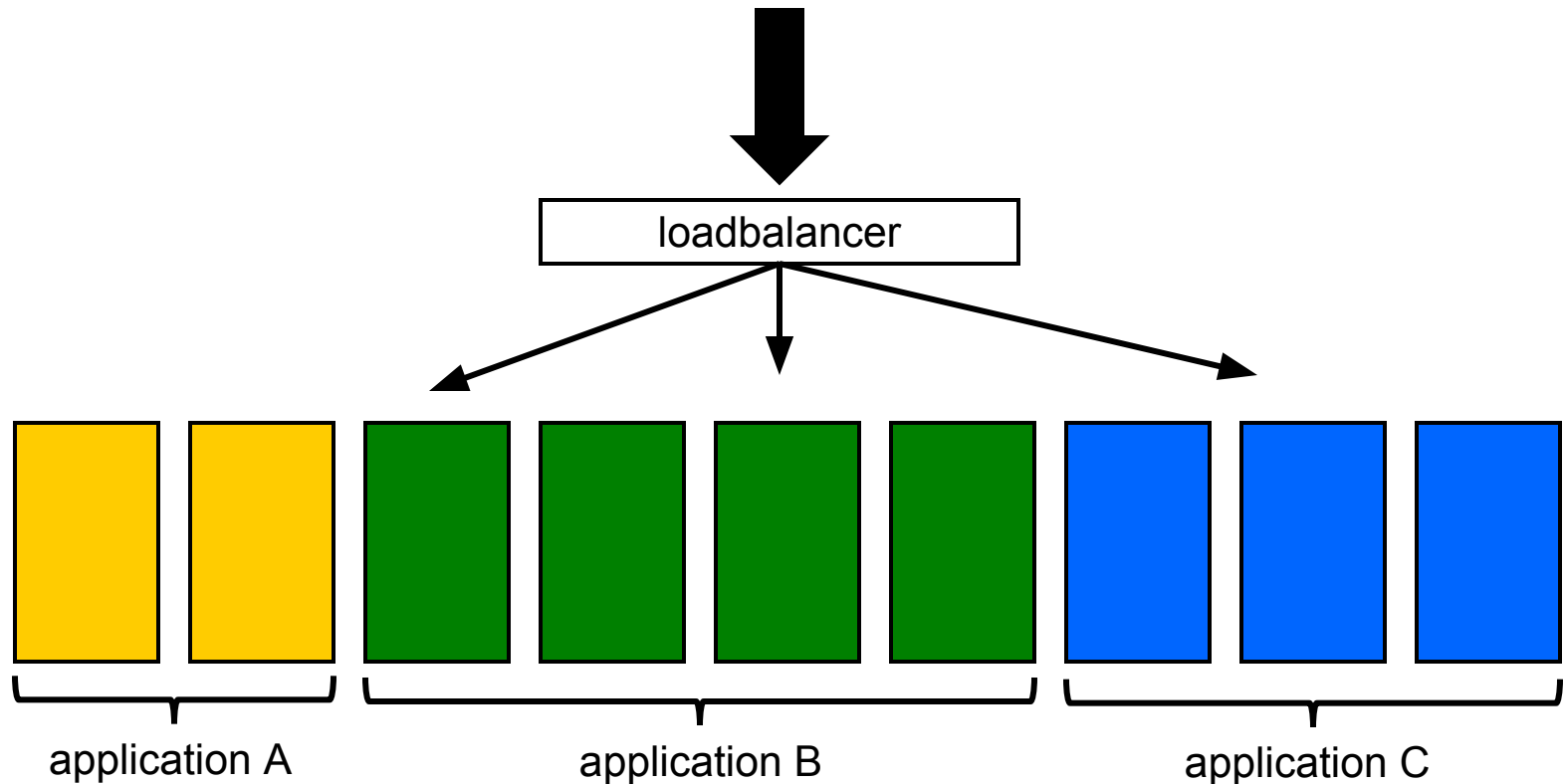
Other examples

- pole-balancing
- TD-Gammon [Gerry Tesauro]
- helicopter [Andrew Ng]



- no teacher who would say “good” or “bad”
 - is reward “10” good or bad?
 - rewards could be delayed
- similar to control theory
 - more general, fewer constraints
- explore the environment and learn from experience
 - not just blind search, try to be smart about it

Resource allocation in datacenters



- A Hybrid Reinforcement Learning Approach to Autonomic Resource Allocation
 - Tesauro, Jong, Das, Bennani (IBM)
 - ICAC 2006

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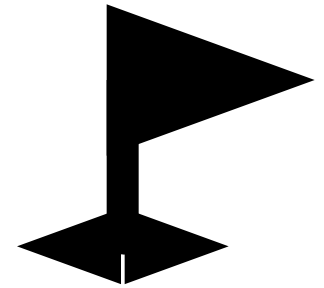
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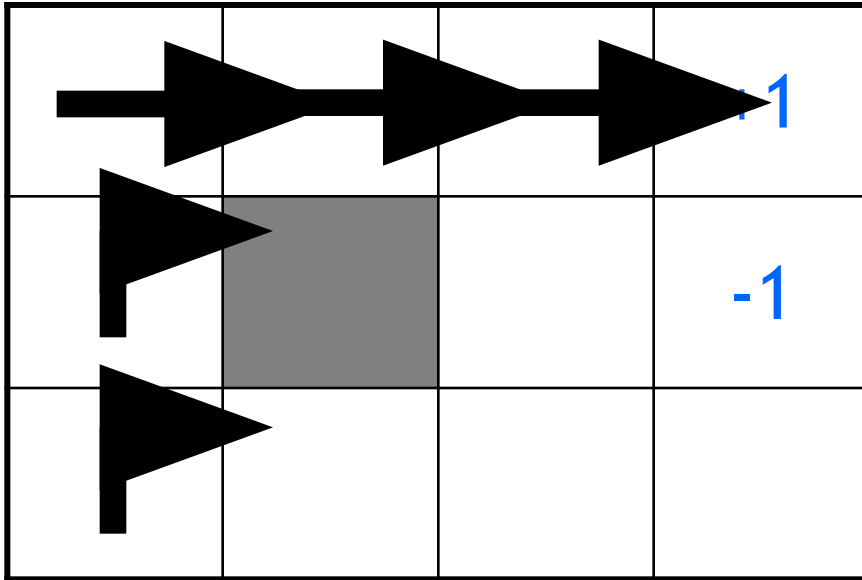
80% move UP
10% move LEFT
10% move RIGHT



reward +1 at [4,3], -1 at [4,2]
reward -0.04 for each step

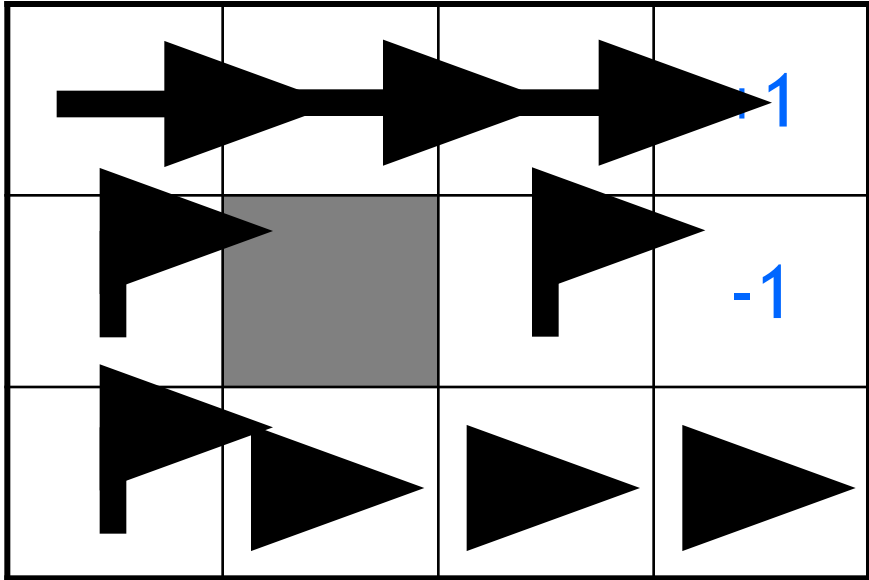
- states
- actions
- rewards
- what is the solution?

Is this a solution?

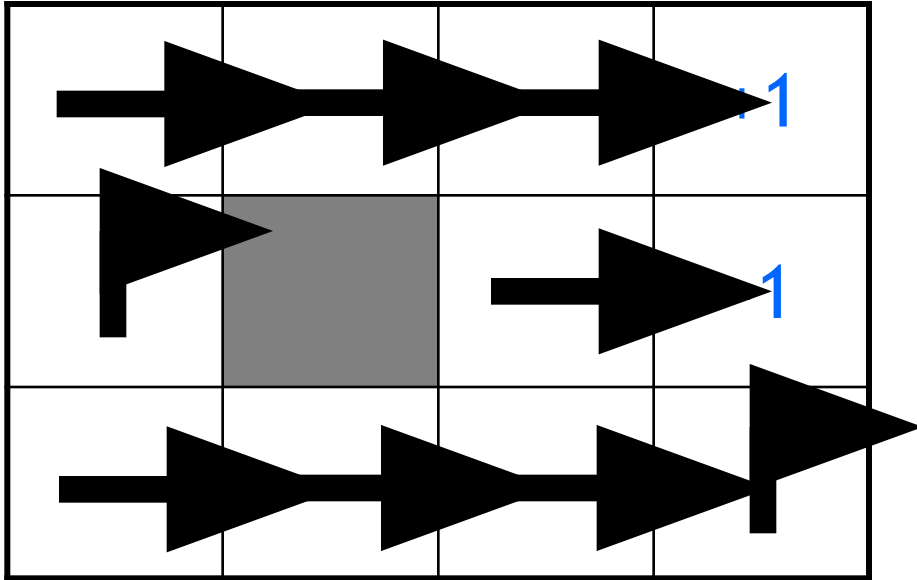


- only if actions deterministic
 - not in this case (actions are stochastic)
- solution/policy
 - mapping from each state to an action

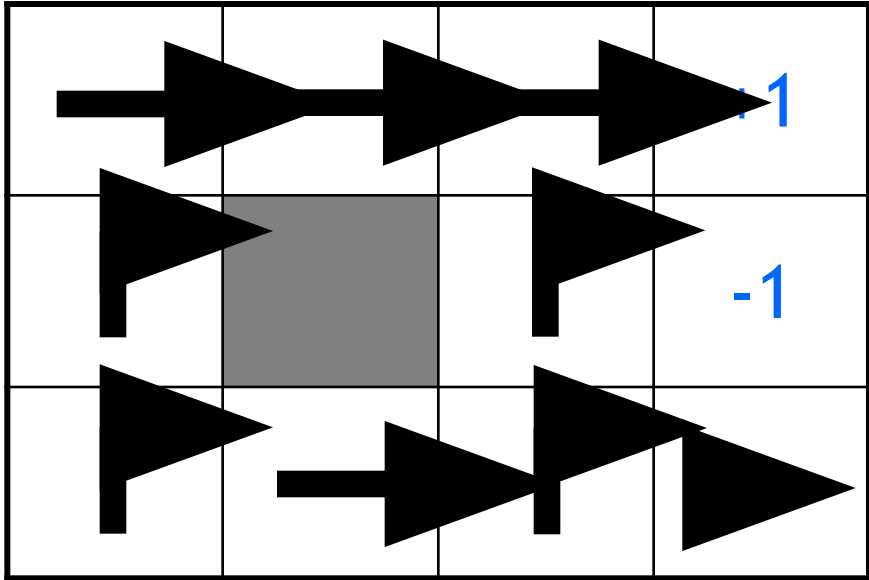
Optimal policy



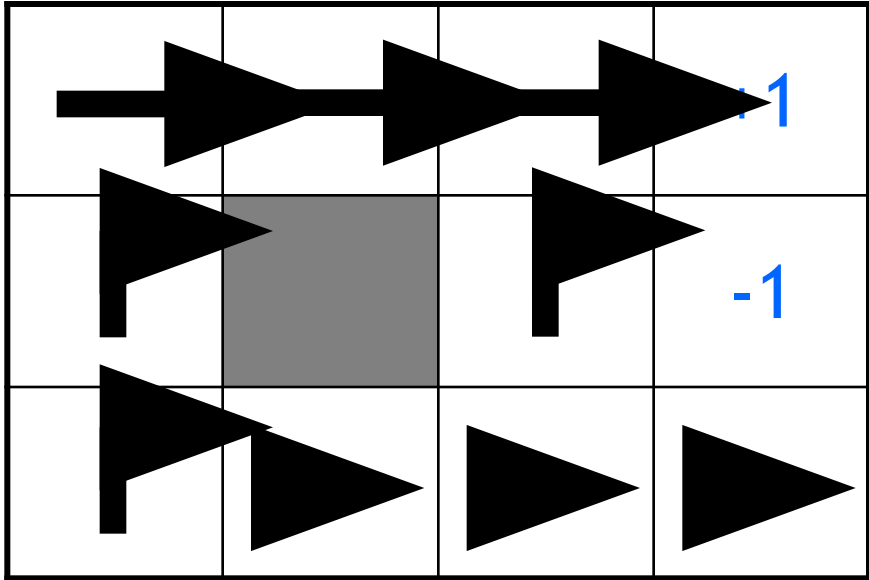
Reward for each step: -2



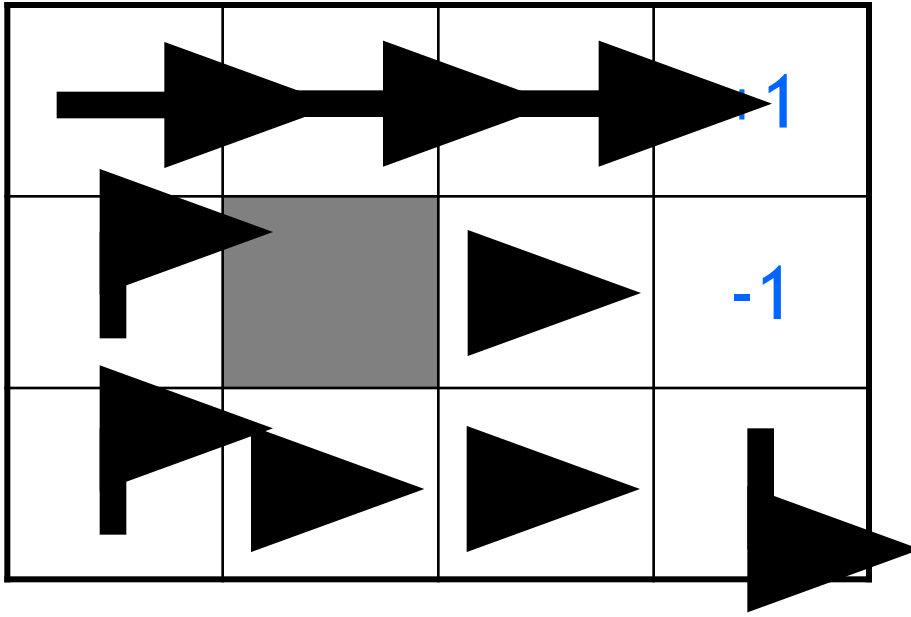
Reward for each step: -0.1



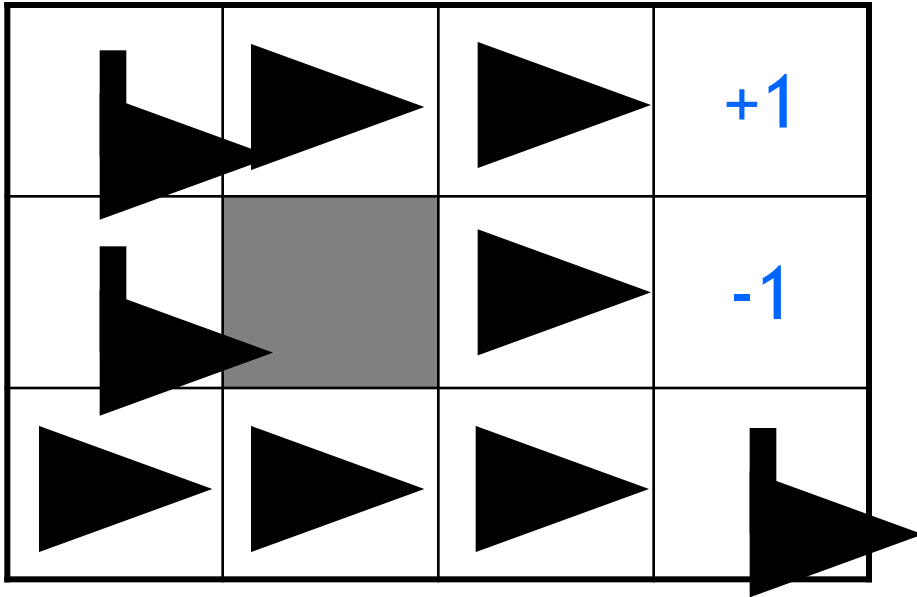
Reward for each step: -0.04



Reward for each step: -0.01

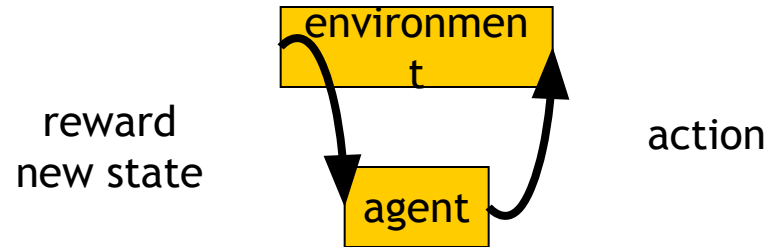


Reward for each step: +0.01



Markov Decision Process (MDP)

- set of states S , set of actions A , initial state S_0
- transition model $P(s,a,s')$
 - $P([1,1], \text{up}, [1,2]) = 0.8$
- reward function $r(s)$
 - $r([4,3]) = +1$
- goal: maximize cumulative reward in the long run
- policy: mapping from S to A
 - $\pi(s)$ or $\pi(s,a)$ (deterministic vs. stochastic)
- reinforcement learning
 - transitions and rewards usually not available
 - how to change the policy based on experience
 - how to explore the environment

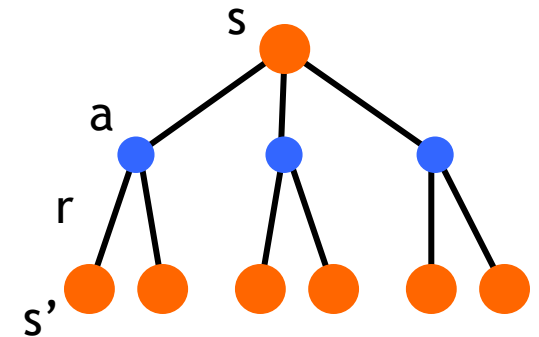


Computing return from rewards

- episodic (vs. continuing) tasks
 - “game over” after N steps
 - optimal policy depends on N; harder to analyze
- additive rewards
 - $V(s_0, s_1, \dots) = r(s_0) + r(s_1) + r(s_2) + \dots$
 - infinite value for continuing tasks
- discounted rewards
 - $V(s_0, s_1, \dots) = r(s_0) + \gamma * r(s_1) + \gamma^2 * r(s_2) + \dots$
 - value bounded if rewards bounded

Value functions

- state value function: $V^\pi(s)$
 - expected return when starting in s and following π
- state-action value function: $Q^\pi(s,a)$
 - expected return when starting in s , performing a , and following π
- useful for finding the optimal policy
 - can estimate from experience
 - pick the best action using $Q^\pi(s,a)$



- Bellman equation

$$V^\pi(s) = \sum_a \pi(s, a) \sum_{s'} P_{ss'}^a [r_{ss'}^a + \gamma V^\pi(s')] = \sum_a \pi(s, a) Q^\pi(s, a)$$

Optimal value functions

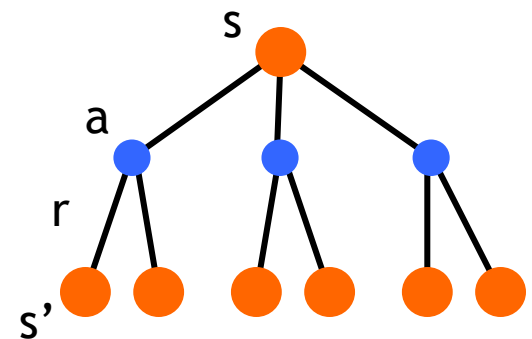
- there's a set of *optimal* policies
 - V^π defines partial ordering on policies
 - they share the same optimal value function

$$V^*(s) = \max_{\pi} V^\pi(s)$$

- Bellman optimality equation

$$V^*(s) = \max_a \sum_{s'} P_{ss'}^a [r_{ss'}^a + \gamma V^*(s')]$$

- system of n non-linear equations
- solve for $V^*(s)$
- easy to extract the optimal policy



- having $Q^*(s,a)$ makes it even simpler

$$\pi^*(s) = \arg \max_a Q^*(s, a)$$

Outline

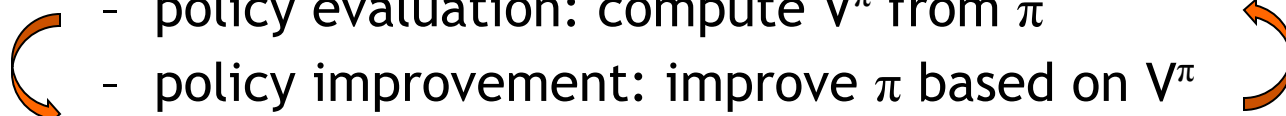
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Dynamic programming

- main idea

- use value functions to structure the search for good policies
- need a perfect model of the environment

- two main components

- policy evaluation: compute V^π from π
 - policy improvement: improve π based on V^π
- 
- start with an arbitrary policy
 - repeat evaluation/improvement until convergence

Policy evaluation/improvement

- policy evaluation: $\pi \rightarrow V^\pi$

- Bellman eqn's define a system of n eqn's
- could solve, but will use iterative version

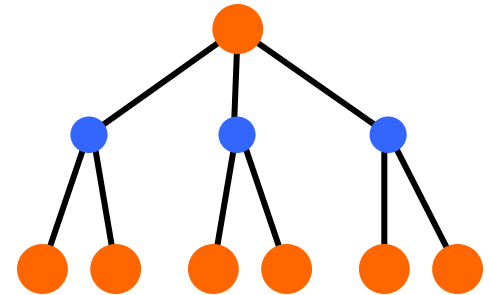
$$V_{k+1}(s) = \sum_a \pi(s, a) \sum_{s'} P_{ss'}^a [r_{ss'}^a + \gamma V_k(s')]$$

- start with an arbitrary value function V_0 , iterate until V_k converges

- policy improvement: $V^\pi \rightarrow \pi'$

$$\begin{aligned} \pi'(s) &= \arg \max_a Q^\pi(s, a) \\ &= \arg \max_a \sum_{s'} P_{ss'}^a [r_{ss'}^a + \gamma V^\pi(s')] \end{aligned}$$

- π' either strictly better than π , or π' is optimal (if $\pi = \pi'$)



Policy/Value iteration

- Policy iteration

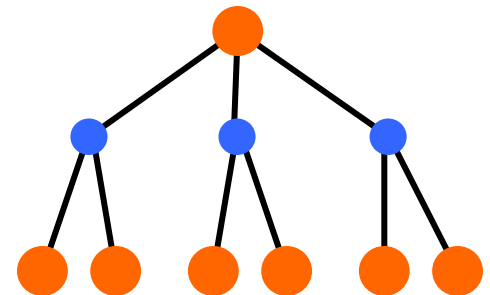
$$\pi_0 \xrightarrow{E} V^{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} V^{\pi_1} \xrightarrow{I} \dots \xrightarrow{I} \pi^* \xrightarrow{E} V^*$$

- two nested iterations; too slow
- don't need to converge to V^{π^k}
 - just move towards it

- Value iteration

$$V_{k+1}(s) = \max_a \sum_{s'} P_{ss'}^a [r_{ss'}^a + \gamma V_k(s')]$$

- use Bellman optimality equation as an update
- converges to V^*



Using DP

- need complete model of the environment and rewards
 - robot in a room
 - state space, action space, transition model
- can we use DP to solve
 - robot in a room?
 - back gammon?
 - helicopter?

Outline

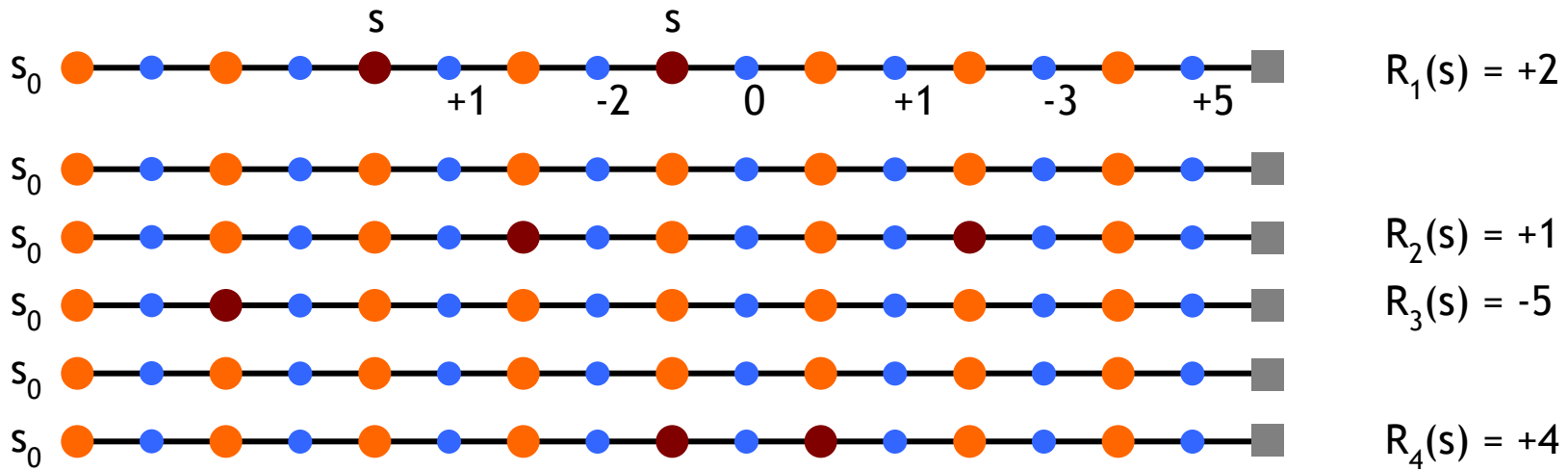
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 - function approximation
 - rewards

Monte Carlo methods

- don't need full knowledge of environment
 - just experience, or
 - simulated experience
- but similar to DP
 - policy evaluation, policy improvement
- averaging sample returns
 - defined only for episodic tasks

Monte Carlo policy evaluation

- want to estimate $V^\pi(s)$
 - = expected return starting from s and following π
 - estimate as average of observed returns in state s
- first-visit MC
 - average returns following the first visit to state s



$$V^\pi(s) \approx (2 + 1 - 5 + 4)/4 = 0.5$$

Monte Carlo control

- V^π not enough for policy improvement
 - need exact model of environment

- estimate $Q^\pi(s,a)$

$$\pi'(s) = \arg \max_a Q^\pi(s, a)$$

- MC control

$$\pi_0 \xrightarrow{E} Q^{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} Q^{\pi_1} \xrightarrow{I} \dots \xrightarrow{I} \pi^* \xrightarrow{E} Q^*$$

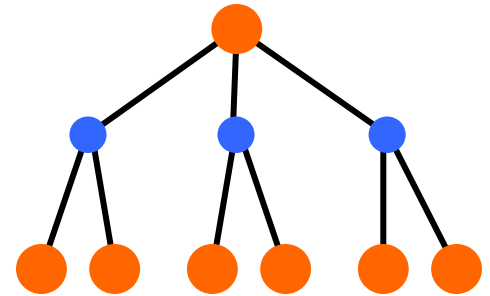
- update after each episode

- non-stationary environment

$$V(s) \leftarrow V(s) + \alpha [R - V(s)]$$

- a problem

- greedy policy won't explore all actions



Maintaining exploration

- deterministic/greedy policy won't explore all actions
 - don't know anything about the environment at the beginning
 - need to try all actions to find the optimal one
- maintain exploration
 - use *soft* policies instead: $\pi(s,a) > 0$ (for all s,a)
- ϵ -greedy policy
 - with probability $1-\epsilon$ perform the optimal/greedy action
 - with probability ϵ perform a random action
 - will keep exploring the environment
 - slowly move it towards greedy policy: $\epsilon \rightarrow 0$

Simulated experience

- 5-card draw poker

- s_0 : A♣, A♦, 6♠, A♥, 2♠
- a_0 : discard 6♠, 2♠
- s_1 : A♣, A♦, A♥, A♠, 9♠ + dealer takes 4 cards
- return: +1 (probably)

- DP

- list all states, actions, compute $P(s,a,s')$
 - $P([A♣, A♦, 6♠, A♥, 2♠], [6♠, 2♠], [A♠, 9♠, 4]) = 0.00192$

- MC

- all you need are sample episodes
- let MC play against a random policy, or itself, or another algorithm

Summary of Monte Carlo

- don't need model of environment
 - averaging of sample returns
 - only for episodic tasks
- learn from sample episodes or simulated experience
- can concentrate on “important” states
 - don't need a full sweep
- need to maintain exploration
 - use soft policies

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Temporal Difference Learning

- combines ideas from MC and DP

- like MC: learn directly from experience (don't need a model)
- like DP: learn from values of successors
- works for continuous tasks, usually faster than MC

- constant-alpha MC:

- have to wait until the end of episode to update

$$V(s_t) \leftarrow V(s_t) + \alpha [R_t - V(s_t)]$$



target

- simplest TD

- update after every step, based on the successor

$$V(s_t) \leftarrow V(s_t) + \alpha [r_{t+1} + \gamma V(s_{t+1}) - V(s_t)]$$



MC vs. TD

- observed the following 8 episodes:

A - 0, B - 0	B - 1	B - 1	B - 1
B - 1	B - 1	B - 1	B - 0

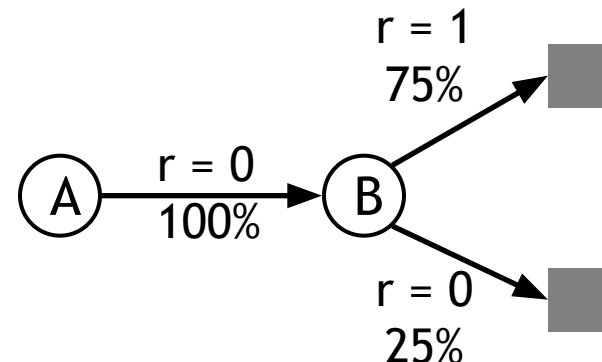
- MC and TD agree on $V(B) = 3/4$

- MC: $V(A) = 0$

- converges to values that minimize the error on training data

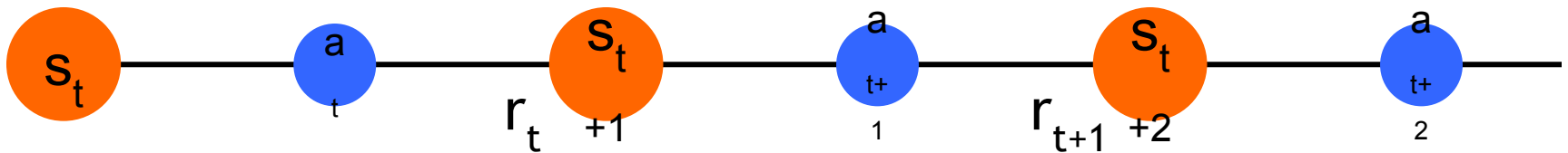
- TD: $V(A) = 3/4$

- converges to ML estimate of the Markov process



Sarsa

- again, need $Q(s,a)$, not just $V(s)$



$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)]$$

- control
 - start with a random policy
 - update Q and π after each step
 - again, need ϵ -soft policies

The RL Intro book

Reinforcement Learning

An Introduction



Richard S. Sutton and Andrew G. Barto

Richard Sutton, Andrew Barto
Reinforcement Learning,
An Introduction

<http://www.cs.ualberta.ca/~sutton/book/the-book.html>

Backup slides

Q-learning

- before: on-policy algorithms
 - start with a random policy, iteratively improve
 - converge to optimal

- Q-learning: off-policy
 - use any policy to estimate Q

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha [r_{t+1} + \gamma \max_a Q(s_{t+1}, a) - Q(s_t, a_t)]$$

- Q directly approximates Q^* (Bellman optimality eqn)
- independent of the policy being followed
- only requirement: keep updating each (s,a) pair

- Sarsa

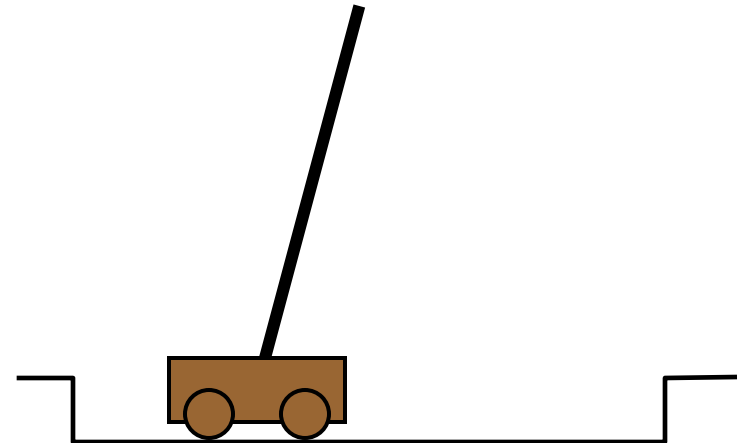
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 - function approximation
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State representation

- pole-balancing
 - move car left/right to keep the pole balanced
- state representation
 - position and velocity of car
 - angle and angular velocity of pole
- what about *Markov property*?
 - would need more info
 - noise in sensors, temperature, bending of pole
- solution
 - coarse discretization of 4 state variables
 - left, center, right
 - totally non-Markov, but still works



Function approximation

- represent V_t as a parameterized function
 - linear regression, decision tree, neural net, ...
 - linear regression: $V_t(s) = \vec{\theta}_t^T \vec{\phi}_s = \sum_{i=1}^n \theta_t(i) \phi_s(i)$
- update parameters instead of entries in a table
 - better generalization
 - fewer parameters and updates affect “similar” states as well

- TD update

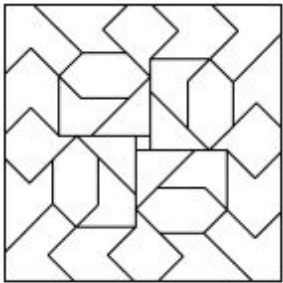
$$V(s_t) \leftarrow V(s_t) + \alpha [r_{t+1} + \gamma V(s_{t+1}) - V(s_t)]$$

$$V(\underbrace{s_t}_{\mathbf{x}}) \mapsto \underbrace{r_{t+1} + \gamma V(s_{t+1})}_{\mathbf{y}}$$

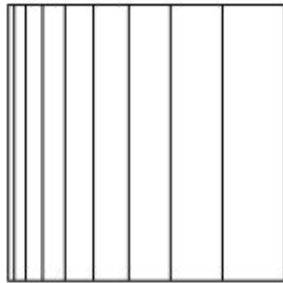
- treat as one data point for regression
- want method that can learn on-line (update after each step)

Features

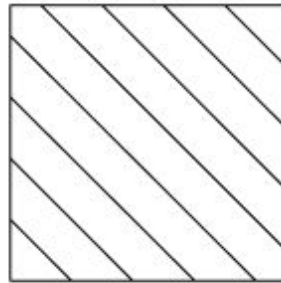
- tile coding, coarse coding
 - binary features



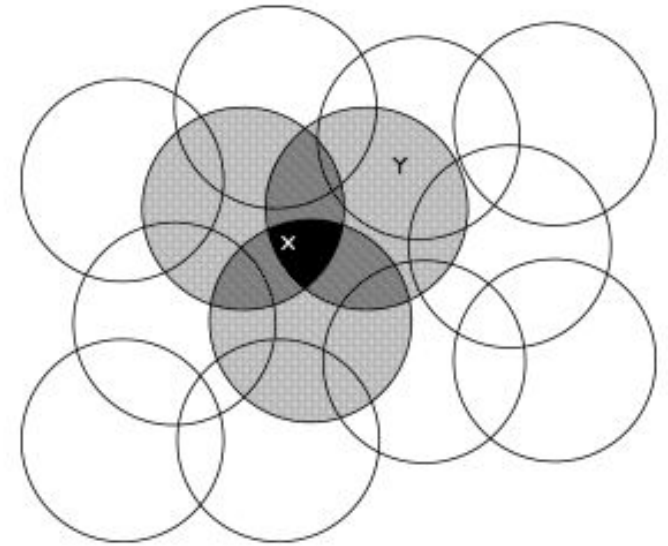
a) Irregular



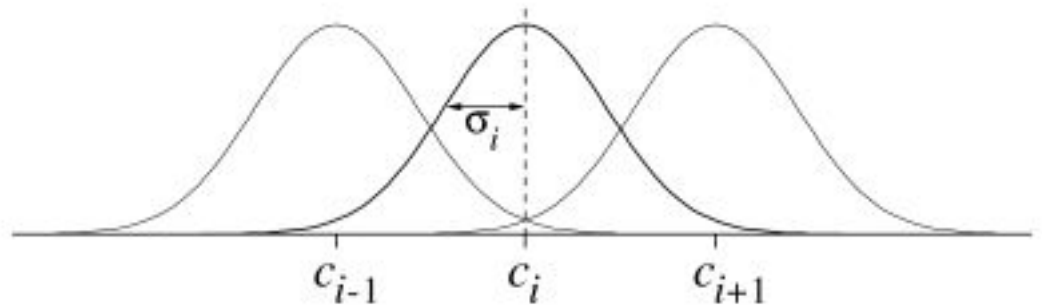
b) Log stripes



c) Diagonal stripes



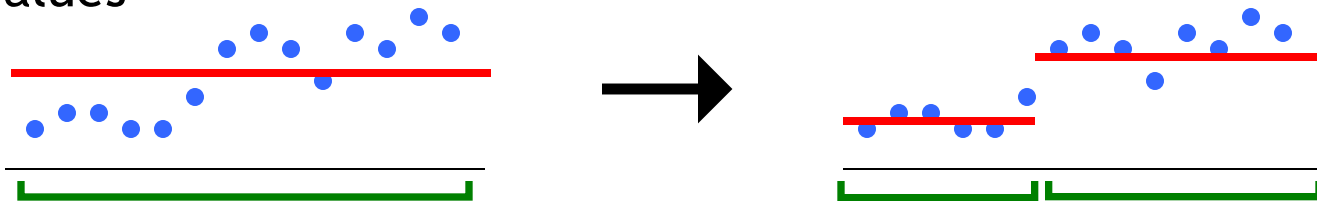
- radial basis functions
 - typically a Gaussian
 - between 0 and 1



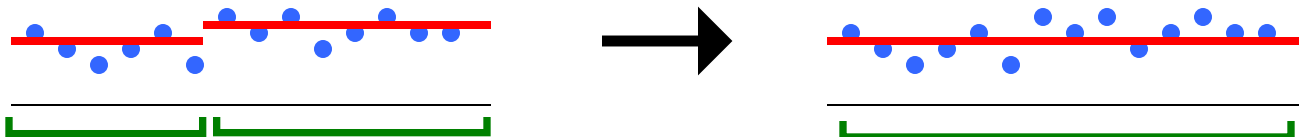
[Sutton & Barto, Reinforcement Learning]

Splitting and aggregation

- want to discretize the state space
 - learn the best discretization during training
- splitting of state space
 - start with a single state
 - split a state when different *parts of that state* have different values



- state aggregation
 - start with many states
 - merge states with similar values



Designing rewards

- robot in a maze

- episodic task, not discounted, +1 when out, 0 for each step

- chess

- GOOD: +1 for winning, -1 losing
- BAD: +0.25 for taking opponent's pieces
 - high reward even when lose

- rewards

- rewards indicate what we want to accomplish
- NOT how we want to accomplish it

- shaping



- positive reward often very “far away”
- rewards for achieving subgoals (domain knowledge)
- also: adjust initial policy or initial value function

Case study: Back gammon

- rules

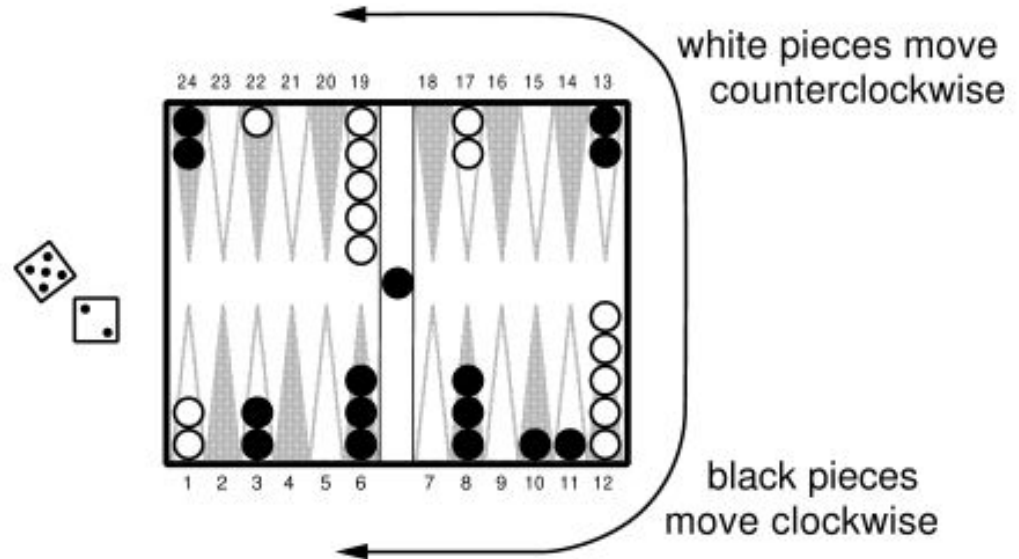
- 30 pieces, 24 locations
- roll 2, 5: move 2, 5
- hitting, blocking
- branching factor: 400

- implementation

- use TD(λ) and neural nets
- 4 binary features for each
- no BG expert knowledge

- results

- TD-Gammon 0.0: trained against itself (300,000 games)
 - as good as best previous BG computer program (also by Tesauro)
 - lot of expert input, hand-crafted features
- TD-Gammon 1.0: add special features
- TD-Gammon 2 and 3 (2-ply and 3-ply search)
 - 1.5M games, beat human champion



Summary

- Reinforcement learning
 - use when need to make decisions in uncertain environment
- solution methods
 - dynamic programming
 - need complete model
 - Monte Carlo
 - time-difference learning (Sarsa, Q-learning)
- most work
 - algorithms simple
 - need to design features, state representation, rewards